Digital Image Processing Lab Assignment 2

Assignment: RGB to Grayscale and Negative Image Conversion

Input Image

Each member of the team must use a separate RGB image.

Tasks

Task 1: RGB to Grayscale Conversion

Convert the RGB image into a grayscale image.
You may use the standard grayscale conversion formula:
Gray=0.299R+0.587G+0.114B
Save the output as:
gray_image.jpg

Task 2: Grayscale to Negative Image

Using the grayscale image obtained in Task 1, generate its negative image using the following formula:
Negative=255−Gray
Save the output as:
Negative_image.jpg

Example Transformation :



Task 3: (You don't have to write code for task 3, but you do have to answer this in the lab.)

Answer the following question:
Can the final negative image be converted back into the original RGB image? Why or why not?

Expected Output

- Source code
- Original RGB image
- Grayscale image
- Negative image